

Diskrete Mathematik – Woche 4

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<https://github.com/tschai-yim/eth-resources>

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Predicate Logic

Problem 1 ★

Let $U = \mathbb{N}$, with the usual meaning of $+$, $\cdot$, $<$, $\leq$, $=$.

1. Determine the truth value of each of the following formulas:

$$F_1 = \forall x((x \geq 5) \rightarrow (x^2 \geq 25)), \quad F_2 = \forall x((x \geq 5) \wedge (x^2 \geq 25)),$$

$$F_3 = \exists x((x \geq 5) \wedge (x^2 \geq 25)), \quad F_4 = \exists x((x \geq 5) \rightarrow (x^2 \geq 25)).$$

2. Which of them formalize “every natural number ≥ 5 has a square ≥ 25 ”, and which formalize “some natural number ≥ 5 has a square ≥ 25 ”?

3. Determine the truth value of F_4 when $x^2 \geq 25$ is replaced by $x^2 < 0$. What does this say about using $\rightarrow$ under $\exists$?

Solution:

1. $F_1 = \top$: for $x < 5$ the premise is $\perp$ and the implication holds, for $x \geq 5$ we get $x^2 \geq 5x \geq 25$.

$F_2 = \perp$: the body must hold for **every** x , but $x = 0$ already fails $x \geq 5$.

$F_3 = \top$: $x = 5$ satisfies both conjuncts.

$F_4 = \top$, but only trivially: $x = 0$ makes the premise $\perp$, so the implication is $\top$.

2. Only F_1 formalizes the $\forall$ -statement, only F_3 the $\exists$ -statement (cf. the notation $\forall x \geq 5 : (x^2 \geq 25)$ in the script).

3. $\exists x((x \geq 5) \rightarrow (x^2 < 0))$ is still $\top$, witnessed by $x = 0$, although no natural number satisfies $x^2 < 0$. An $\exists$ combined with $\rightarrow$ is satisfied by every element violating the premise, so it says almost nothing. Restriction under $\exists$ must use $\wedge$, under $\forall$ it must use $\rightarrow$.

Problem 2 ★ ★

Decide for each of the following formulas for which of the universes $U = \mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ it is true ($+$, $\cdot$, $<$, $\leq$, $=$ with their usual meaning).

1. $G_1 = \forall x \exists y (y + y = x)$

2. $G_2 = \forall x ((x > 0) \rightarrow \exists y (y \cdot y = x))$

3. $G_3 = \forall x \exists y ((x < y) \wedge \forall z (\neg((x < z) \wedge (z < y))))$

4. $G_4 = \exists x \forall y (x \leq y)$

Solution:

	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$
G_1	$\perp$	$\perp$	$\top$	$\top$
G_2	$\perp$	$\perp$	$\perp$	$\top$
G_3	$\top$	$\top$	$\perp$	$\perp$
G_4	$\top$	$\perp$	$\perp$	$\perp$

1. “Every element is halvable”. Fails for $x = 1$ in $\mathbb{N}$ and $\mathbb{Z}$; in $\mathbb{Q}$ and $\mathbb{R}$ take $y = \frac{x}{2}$.
2. “Every positive element is a square”. $x = 2$ is a counterexample in $\mathbb{N}$, $\mathbb{Z}$ and $\mathbb{Q}$ ($\sqrt{2} \notin \mathbb{Q}$); in $\mathbb{R}$ take $y = \sqrt{x}$.
3. “Every element has an immediate successor”, i.e. some $y > x$ with no z strictly in between. Take $y = x + 1$ in $\mathbb{N}$ and $\mathbb{Z}$. In $\mathbb{Q}$ and $\mathbb{R}$ it fails: for any $y > x$ the element $z = \frac{x+y}{2}$ lies strictly in between.
4. “There is a smallest element”. True in $\mathbb{N}$ with $x = 0$; in $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ every x has the smaller element $x - 1$.

Note the order of quantifiers in 4.: the x must be chosen **before** y , so one single x has to work for all y .

Problem 3 ★ ★

Let $U = \mathbb{N} \setminus \{0\}$. The only symbols you may use are the binary predicate $D(x, y) = 1 \Leftrightarrow “x \text{ divides } y”$, equality $=$, and the logical operators and quantifiers. Express the following statements as formulas (with free variables x, y, z as indicated):

1. “ x is prime”
2. “ x and y are coprime”, i.e. their only common divisor is 1
3. “ z is the greatest common divisor of x and y ” (★ ★ ★)

Solution:

1. $\text{prime}(x) \Leftrightarrow \neg(x = 1) \wedge \forall z(D(z, x) \rightarrow ((z = 1) \vee (z = x)))$. The clause $\neg(x = 1)$ is necessary: 1 satisfies the divisor condition. Compare with Example 2.19, where $\text{prime}(x)$ is written with multiplication instead.
2. $\text{coprime}(x, y) \Leftrightarrow \forall z((D(z, x) \wedge D(z, y)) \rightarrow (z = 1))$.
3. $\text{gcd}(x, y, z) \Leftrightarrow D(z, x) \wedge D(z, y) \wedge \forall w((D(w, x) \wedge D(w, y)) \rightarrow D(w, z))$.

“Greatest” cannot be written as $w \leq z$, since no order predicate is available. Instead maximality is expressed via divisibility: z is a common divisor that every common divisor divides. On $\mathbb{N} \setminus \{0\}$ this gives the same z , because $D(w, z)$ implies $w \leq z$.

Problem 4 ★ ★ ★

As in the script (Sections 2.4.6–2.4.7), an informal justification suffices here; interpretations are only defined rigorously in Chapter 6.

1. Prove $\exists y \forall x (P(x, y)) \models \forall x \exists y (P(x, y))$, and disprove $\forall x \exists y (P(x, y)) \models \exists y \forall x (P(x, y))$ by giving an interpretation.
2. Is $\exists x (P(x) \rightarrow \forall y (P(y)))$ valid, i.e. true under every interpretation? Prove it or give an interpretation in which it is false.

3. Is $\exists y \forall x (Q(x, y) \leftrightarrow \neg Q(x, x))$ satisfiable?

Solution:

1. **Consequence holds.** Let the premise be $\top$ under some interpretation: there is an $a \in U$ with $P(x, a) = \top$ for all $x \in U$. Then for every x the element $y := a$ witnesses $\exists y (P(x, y))$, so the conclusion is $\top$. **Converse fails.** Let $U = \mathbb{Z}$ and $P(x, y) = 1 \Leftrightarrow y < x$. The premise is $\top$ (take $y = x - 1$), but the conclusion is $\perp$: no single integer is smaller than all integers. Hence $\forall x \exists y (P(x, y)) \not\models \exists y \forall x (P(x, y))$.
2. **Valid.** Let an arbitrary interpretation be given and distinguish two cases (Definition 2.16):
 - $\forall y (P(y))$ is $\top$: pick any $a \in U$ (possible, since $U \neq \emptyset$). The implication has a true conclusion, so $x := a$ works.
 - $\forall y (P(y))$ is $\perp$: then $P(a) = \perp$ for some $a \in U$. The implication has a false premise, so $x := a$ works.

In both cases some x satisfies the body, so the formula is $\top$. Note that the non-emptiness of U is used.

3. **Unsatisfiable.** Assume it is $\top$ under some interpretation: there is $b \in U$ such that $Q(x, b) \leftrightarrow \neg Q(x, x)$ is $\top$ for **all** $x \in U$. Choosing $x := b$ gives $Q(b, b) \leftrightarrow \neg Q(b, b)$, which is $\perp$ for either value of $Q(b, b)$ – a contradiction.

Proof Techniques

Problem 5 ★ ★

In the script every proof pattern is justified by a lemma of propositional logic (Lemmas 2.5–2.9), where A stands for “ S is true” and B for “ T is true”. For each of the following patterns, write down the corresponding statement about formulas and then **prove** or **disprove** it.

1. To prove $S \Rightarrow T$: assume S and that T is false, and derive a contradiction.
2. To prove $S \Rightarrow T$: prove $\neg S \Rightarrow \neg T$.
3. To prove S : find a statement T and prove both $T \Rightarrow S$ and $\neg T \Rightarrow S$.

Solution:

1. **Sound.** The pattern shows that $A \wedge \neg B$ is unsatisfiable, and claims $\models A \rightarrow B$:

$$\begin{aligned} A \wedge \neg B \equiv \perp &\Leftrightarrow \neg(A \wedge \neg B) \text{ is a tautology} && \text{(Lemma 2.2)} \\ &\Leftrightarrow \neg A \vee B \text{ is a tautology} && \text{(De Morgan, double negation)} \\ &\Leftrightarrow A \rightarrow B \text{ is a tautology} && \text{(Implication equivalence)} \end{aligned}$$

and $\models A \rightarrow B$ is the same as $A \models B$ (Lemma 2.3).

2. **Unsound.** The pattern claims $\neg A \rightarrow \neg B \models A \rightarrow B$. Counterexample: the truth assignment $A = 1, B = 0$ makes $\neg A \rightarrow \neg B \equiv 0 \rightarrow 1$ true, while $A \rightarrow B$ is false. The pattern confuses the **converse** with the **contraposition**: $\neg A \rightarrow \neg B \equiv B \rightarrow A$, whereas the indirect proof (Definition 2.14, Lemma 2.6) uses $\neg B \rightarrow \neg A \equiv A \rightarrow B$.
3. **Sound.** The pattern claims $(A \rightarrow B) \wedge (\neg A \rightarrow B) \models B$. This is Lemma 2.8 (case distinction) for $k = 2$ with $A_1 = A$ and $A_2 = \neg A$, where step 2 of Definition 2.16 is free because $A \vee \neg A \equiv \top$. Directly: if $B = 0$ under some assignment, then $A = 1$ falsifies $A \rightarrow B$ and $A = 0$ falsifies $\neg A \rightarrow B$, so the premise is never true when B is false.

Problem 6 ★ ★

Each of the following “proofs” is flawed. Identify the invalid or unjustified step, and state whether the claim itself is true or false.

1. **Claim:** For all $n \in \mathbb{N}$: if n^2 is even, then n is even.
“Proof”: Let n be even, i.e. $n = 2k$. Then $n^2 = 4k^2 = 2(2k^2)$ is even. ■
2. **Claim:** For all $x \in \mathbb{R}$: $x^2 \geq x$.
“Proof”: Case $x \leq 0$: then $x^2 \geq 0 \geq x$. Case $x \geq 1$: then $x^2 = x \cdot x \geq 1 \cdot x = x$. Both cases hold. ■
3. **Claim:** There are infinitely many primes.
“Proof”: Assume $p_1, \dots, p_n$ are all primes. Set $P = p_1 \cdot \dots \cdot p_n + 1$. No p_i divides P , hence P is prime. But $P > p_i$ for all i , contradicting that $p_1, \dots, p_n$ are all primes. ■

Solution:

1. The “proof” shows $n \text{ even} \Rightarrow n^2 \text{ even}$, i.e. the **converse** of the claim. The claim is **true**, but needs an indirect proof (Definition 2.14): $n \text{ odd} \Rightarrow n = 2k + 1 \Rightarrow n^2 = 2(2k^2 + 2k) + 1$ odd.
2. The cases are not exhaustive: $0 < x < 1$ is covered by neither, so step 2 of Definition 2.16 (“at least one case occurs”) fails. The claim is **false**: $x = \frac{1}{2}$ gives $x^2 = \frac{1}{4} < \frac{1}{2}$.
3. The step “no p_i divides P , hence P is prime” is unjustified — avoiding the p_i as divisors does not make a number prime. The claim is **true**, and the gap closes by using the known fact that every $n \geq 2$ has a prime divisor q ; here $q \notin \{p_1, \dots, p_n\}$ gives the contradiction (cf. Example 2.30).

Problem 7 ★ ★

1. Prove indirectly (Definition 2.14): for all $x, y \in \mathbb{R}$, if $x \cdot y$ is irrational, then x is irrational or y is irrational.
2. Decide whether the converse holds: if x is irrational or y is irrational, then $x \cdot y$ is irrational.

Solution:

1. Negating both sides and applying de Morgan to the conclusion gives the statement to be proven:

if x and y are rational, then $x \cdot y$ is rational.

Let $x = \frac{a}{b}$ and $y = \frac{c}{d}$ with $a, c \in \mathbb{Z}$ and $b, d \in \mathbb{Z} \setminus \{0\}$. Then $x \cdot y = \frac{ac}{bd}$ with $bd \neq 0$, hence $x \cdot y$ is rational. ■

2. **False.** Take $x = y = \sqrt{2}$, which is irrational by Example 2.28; then $x \cdot y = 2$ is rational. (Even simpler: $x = \sqrt{2}, y = 0$.)

Problem 8 ★ ★

1. Show by contradiction (Definition 2.17) that there is no smallest positive rational number, i.e. that for $U = \mathbb{Q}$

$$\neg \exists q(q > 0 \wedge \forall r((r > 0) \rightarrow (q \leq r))).$$

2. The analogous statement is false for $U = \mathbb{N}$. Which step of your proof breaks, and which property of $\mathbb{Q}$ does the proof really use?

Solution:

1. Assume the statement is false, i.e. such a q exists. Consider $\frac{q}{2}$:

- $\frac{q}{2}$ is rational, since $\mathbb{Q}$ is closed under division by 2,
- $\frac{q}{2} > 0$, since $q > 0$,
- $\frac{q}{2} < q$, since $q > 0$.

Instantiating the minimality of q with $r := \frac{q}{2}$ yields $q \leq \frac{q}{2}$, hence $q \leq 0$ – which contradicts $q > 0$. ■

2. The step “ $\frac{q}{2}$ lies in U ” breaks: for $q = 1 \in \mathbb{N}$ we have $\frac{1}{2} \notin \mathbb{N}$, and indeed 1 is the smallest positive natural number. The proof does not use the ordering alone but the closure of $\mathbb{Q}$ under halving (cf. Example 2.18, where $\forall x \exists y(y < x)$ separates $\mathbb{N}$ from $\mathbb{Z}$ and $\mathbb{R}$).

Problem 9 ★ ★ ★

Show that there exist irrational numbers a, b such that a^b is rational. You may use that $\sqrt{2}$ is irrational (Example 2.28) and that $(x^y)^z = x^{y \cdot z}$ for $x > 0$. Afterwards decide whether your proof is constructive or non-constructive (Definition 2.18).

Solution:

Case distinction on the number $\sqrt{2}^{\sqrt{2}}$, which is either rational or irrational:

- **Case 1:** $\sqrt{2}^{\sqrt{2}}$ is **rational**. Take $a = b = \sqrt{2}$, both irrational. Then $a^b = \sqrt{2}^{\sqrt{2}}$ is rational. ✓
- **Case 2:** $\sqrt{2}^{\sqrt{2}}$ is **irrational**. Take $a = \sqrt{2}^{\sqrt{2}}$ and $b = \sqrt{2}$, both irrational. Then

$$a^b = \left(\sqrt{2}^{\sqrt{2}} \right)^{\sqrt{2}} = \sqrt{2}^{\sqrt{2} \cdot \sqrt{2}} = \sqrt{2}^2 = 2. \quad \checkmark$$

At least one of the two cases occurs and in both a suitable pair exists, which proves the existence statement. ■

The proof is **non-constructive**: it never decides which case applies, so no concrete pair (a, b) is exhibited — only the guarantee that one of the two listed pairs works.